

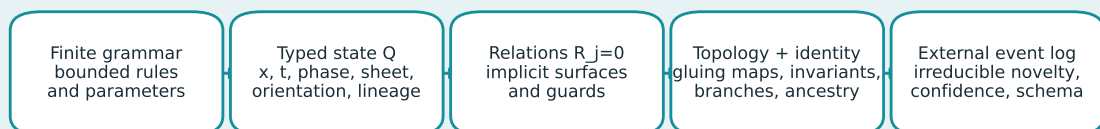
Unified Geometric-Topological Substrate

A clean information architecture extracted from the supplied geometry, game-technology, graphics and topology corpus

UGTS-0 architecture

Authoritative relation/event substrate with optional projection and hardware adapters

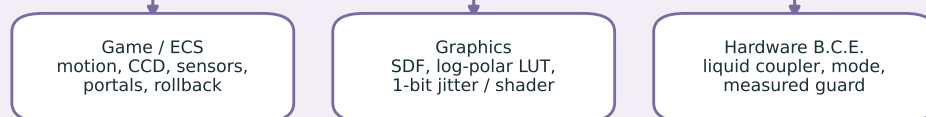
AUTHORITATIVE SUBSTRATE



QUERY / RUNTIME



ADAPTERS



Projection is optional and non-authoritative. One-bit values are schema-bound flags, not complete state.

Deliverable: technical synthesis, full concept inventory, claims ledger, formal specification, reference implementation, tests, examples, graphics shader, game-engine adapters and bounded optofluidic endpoint.

Prepared from nine user-supplied PDFs. Personal identifiers were intentionally omitted from this synthesis.

Version 1.0 - August 2026

Abstract

The supplied documents contain a recurring technical line beneath a large amount of metaphor and speculative extrapolation. The recoverable line is not a new rasterizer and not a claim that all reality is literally spherical. It is a *query-first state-and-event substrate*: a finite grammar produces typed continuous state; local radial-angular supports select relevance; compatibility predicates decide whether co-located sectors may interact; relation surfaces define events; transitions update phase, sheet, orientation, branch and lineage; and an external event log records novelty that cannot be recomputed. Images, game-engine frames and optical devices are downstream adapters.

This report extracts 69 individual mechanisms, assigns each a disposition, and unifies the retained operators into **UGTS-0 / Equation World Zero**. The accompanying Python package implements typed state, linear/quadratic trajectories, implicit geometry, analytic and bracketed event solvers, radial-angular support, phase/sheet compatibility, Mobius and Klein quotient maps, four-chamber hourglass routing, log-polar bit lookup, finite grammars, optional 1-bit projection, and a measured Bounded Compatibility Event controller. Seventeen executable tests pass.

DECISION

The technically coherent core is: **finite grammar + typed state + local support + compatibility + event solving + transition + lineage + external log**. The strongest prototype is intentionally restricted. Universal $O(1)$, zero-memory, zero-latency, physical Klein-bottle self-assembly, and one-bit-complete-state claims are rejected.

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PART 1

Executive unification

1.1 The decisive shift

SOURCE-DERIVED

The mature synthesis in S6 says the archive should no longer be judged primarily as rasterization, raymarching, voxel storage or a frame-loop proposal. Cones, SDF-zero, one-bit, double vacuum and hourglass are recast as operators in a shared state space. Spherical structure is local support, shelling and radial reach, not a universal replacement for every world representation (S6, pp. 1-7).

ENGINEERING NORMALIZATION

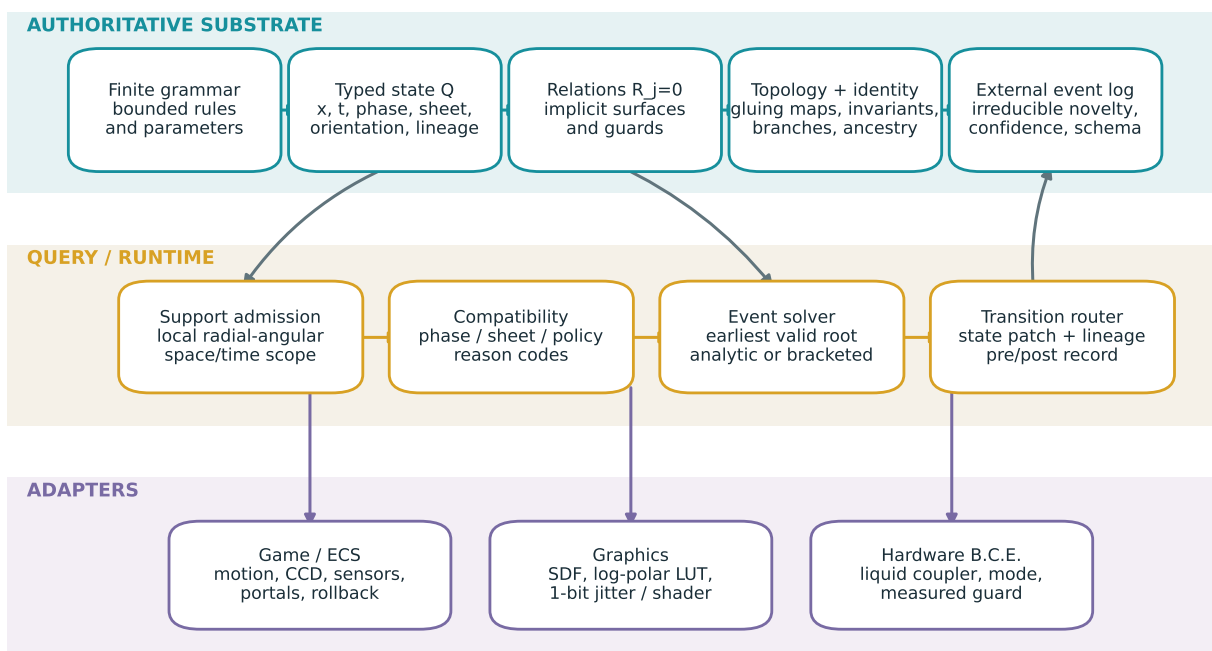
UGTS-0 therefore treats the authoritative object as a relation/event world. A renderer may visualize the world, but the image is not the state. A game loop may schedule queries, but the loop is not the ontology. A one-bit field may route a transition, but it is not the amplitude, uncertainty, identity or geometry.

The unified statement is:

A finite grammar generates a typed continuous state manifold. Local radial-angular supports define relevance and measurement. Events occur when a relation crosses an explicit transition surface while support and compatibility hold. Identity persists by lineage and invariants rather than by instantaneous coordinates. Projection into images, engine objects or photonic outputs is optional and non-authoritative.

UGTS-0 architecture

Authoritative relation/event substrate with optional projection and hardware adapters



Projection is optional and non-authoritative. One-bit values are schema-bound flags, not complete state.

Figure 1.1: UGTS-0 separates authoritative substrate, query runtime and adapters.

1.2 What “works technically” means here

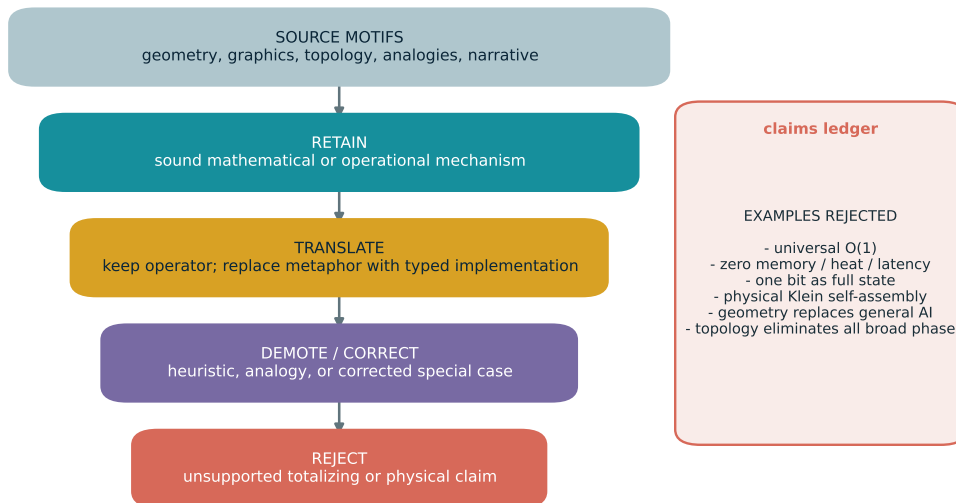
The package does not claim to solve every scene, PDE, physics problem or AI task. “Works” means the following bounded capabilities are executable and tested:

1. Evaluate closed-form entity state at an arbitrary continuous time.
2. Find the earliest valid crossing for supported surface/trajectory families.
3. Filter candidates by local radial-angular support and typed compatibility.
4. Apply explicit state patches while preserving pre-state, post-state and lineage.
5. Represent co-location without coupling by phase/sheet/orientation mismatch.
6. Apply Mobius, Klein, portal and hourglass maps as explicit software gluing/routing.
7. Use log-polar coordinates and a compact one-bit LUT as an optional local address mask.
8. Project implicit fields through an ordinary graphics adapter and posterize to one bit.
9. Run a measured B.C.E. threshold/compatibility controller with a narrow parity bit.

1.3 Retention rule

Evidence and claim filter

Every source motif receives a disposition before entering the technical substrate



Rule: source-derived content and engineering normalization remain visibly distinct in the report, inventory and code comments.

Figure 1.2: The report preserves useful operators while correcting or rejecting unsupported claims.

Each item is marked **RETAIN**, **TRANSLATE**, **DEMOTE** / **CORRECT**, or **REJECT**. Source-derived statements and engineering normalization are kept separate throughout the inventory and code comments.

PART 2

Method and source architecture

2.1 Corpus handling

Nine PDFs were read as design records. Repeated dialogue bundles were deduplicated: S2, S8 and S9 share much of the D/B/phi-to-R thread, so repeated text is not counted as independent validation. S6 and S5 are given the greatest interpretive weight because they explicitly impose feasibility, type-safety, calibration and falsifiability constraints on earlier material.

Table 2.1: Source corpus and role in the extraction. File names are concise aliases; source IDs and page ranges are authoritative within this package.

Ref	Short name	File alias	Role
S1	NUM19 30 pages	numeric-fractal-dialogue.pdf	Numeric, phonetic, binary, triangle, Pascal/Sierpinski, graphics and topology motif chain.
S2	AUTHOR_DIALOGUE 52 pages	author-dialogue-bundle-[identifiers-omitted].pdf	Extended dialogue bundle; duplicates the D/B/phi-to-R thread and adds AR, cognition, networking, and narrative extrapolations.
S3	KC_GRAPHICS 10 pages	kc-vector-graphics.pdf	Graphics adapter: log-polar mapping, SDF/CSG, phasor-style coverage, 1-bit temporal/spatial modulation, chromatic offsets, prepress.
S4	BEN_BURGER 19 pages	ben-burger-zero-index-dialogue.pdf	Zero-based indexing, split-oval/torque wave motif, overlapping semicircles, shared-domain interpretation.
S5	WAVEGUIDE 12 pages	spherical-throughput-waveguide.pdf	Bounded optofluidic hardware translation, B.C.E. logic, metrics, fabrication and kill criteria.
S6	CHRONO_SYNTHESIS 12 pages	chronological-spherical-substrate-synthesis.pdf	Mature internal synthesis: query-first event calculus, local spherical support, compatibility, lineage, feasibility discipline.
S7	HOLLOWLAND 36 pages	hollowland-double-vacuum.pdf	Double-vacuum, phase sheets, analytic cones, hourglass routing, procedural grammar, kinematic calculus and highly speculative extensions.
S8	GEORGE_BUNDLE 42 pages	george-dialogue-bundle.pdf	Substantially overlapping D/B/phi-to-R dialogue bundle; later pages add perception, storage and related extrapolations.
S9	D_TO_PH 34 pages	d-to-ph-glyph-topology.pdf	Core glyph morph, parity hinge, log-polar/L-system, SDF-global hinge and topology motif thread.

2.2 Extraction dimensions

Every candidate mechanism was tested against five questions:

1. **Geometry:** Is there a definable curve, field, support, transform or invariant?
2. **Topology:** Is there an explicit gluing, orientation, sheet or routing operation?
3. **Information architecture:** What state, schema, query, event and provenance are required?
4. **Game technology:** Can the mechanism be expressed as a component, system or adapter?
5. **Evidence boundary:** Is the statement exact, heuristic, speculative or contradicted by ordinary engineering constraints?

2.3 Deduplication and personal-data discipline

The source corpus includes filenames and dialogue text containing personal identifiers. They are not necessary to the geometrical synthesis and are omitted. The package does not redistribute

the original PDFs. Source traceability is by S1-S9, page ranges and descriptive filenames only.

PART 3

Numeric and fractal seeds

3.1 Radix carry thresholds

RETAIN In base b , a new digit is required at b^k . This is an exact positional-number fact and provides a useful scale/level boundary:

$$\text{digits}_b(n) = \begin{cases} 1, & n = 0, \\ \lfloor \log_b n \rfloor + 1, & n > 0. \end{cases}$$

The source begins with binary transitions $1 \rightarrow 10$, $11 \rightarrow 100$, and the decimal transition $9 \rightarrow 10$ (S1, pp. 1–4). UGTS-0 uses radix powers as compact hierarchy thresholds, never as proof that geometry is inherently binary.

3.2 Zero-based offset semantics

RETAIN S4 correctly distinguishes an index from an ordinal: index 35 is the 36th position. The arithmetic $21 + 7 + 7 = 35$ is unchanged. The implementation exposes `zero_based_ordinal(35) == 36` to prevent the narrative “shift” from becoming an off-by-one ambiguity.

3.3 Active bits and the triangle seed

TRANSLATE $19 = 10011_2$ has active positions $\{4, 1, 0\}$ and Hamming weight three. The source maps those three active positions to three Dutch syllables and then to a triangle (S1, pp. 3–7). The technically valid core is popcount and a chosen simplex embedding. The syllable equality is a mnemonic; it is not a general number-language code.

3.4 Pascal parity and Sierpinski

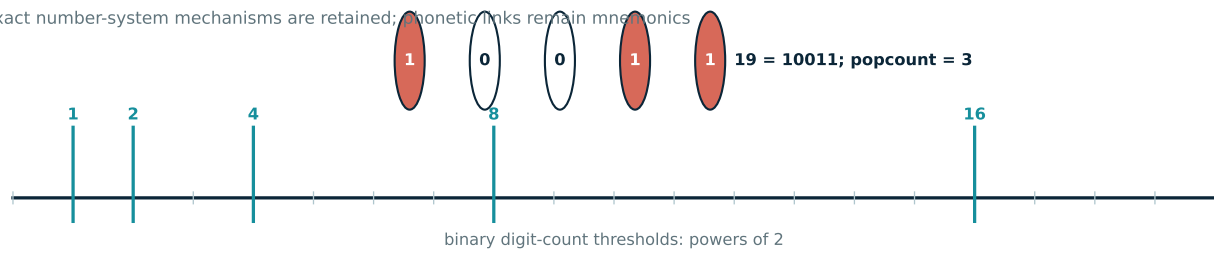
RETAIN Binomial parity is a deterministic fractal generator:

$$\binom{n}{k} \equiv 1 \pmod{2} \iff k \& \neg n = 0.$$

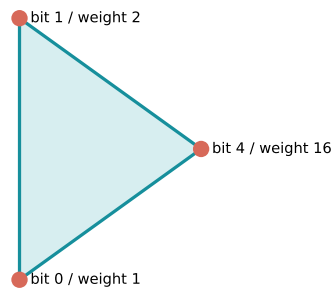
This follows from Lucas-style bit containment. The source’s visual connection is valid, with one correction: zero-based rows $2^m - 1$ are all odd; generic rows 2^m are not. The package contains `pascal_entry_is_odd` and a PBM generator.

Numeric seeds: radix thresholds, active bits and Pascal parity

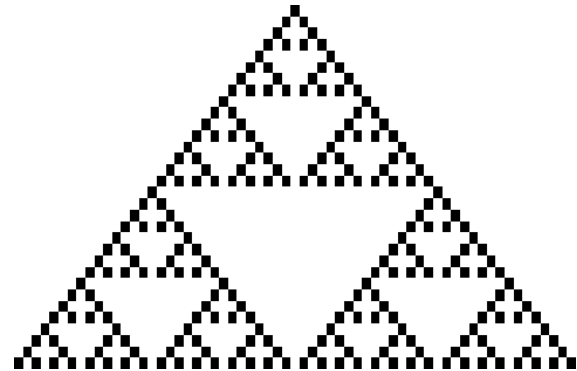
Exact number-system mechanisms are retained; phonetic links remain mnemonics



Chosen triangle embedding of three active positions



Pascal odd/even parity -> Sierpinski test pattern



Correction: zero-based rows $2^k - 1$ are all odd, not generic rows 2^k .

Figure 3.1: Exact numeric mechanisms retained from S1 and S4.

3.5 Golden-angle sampling and strange loops

TRANSLATE Vogel/golden-angle points are useful as a low-regularity sample schedule. They can reduce obvious repetition but do not guarantee zero aliasing or zero cost. **DEMOTE** Lemniscates and Escher loops are retained only as recursion motifs. In UGTS-0 recursion is finite, budgeted and cycle-detectable through FiniteGrammar; no self-creating physical substrate is asserted.

PART 4

Glyph geometry and kinematic tricks

4.1 Loop-plus-stem decomposition

S2, S8 and S9 repeatedly decompose a selected glyph form into a vertical stem plus one or more closed arcs. This depends on the font and therefore becomes a design convention, not a theorem about letters.

4.2 Cut/unroll D or B into R

RETAIN The clearest individual geometrical trick is a boundary-graph edit:

1. Keep the vertical stem and upper loop.
 2. Cut the lower closed arc at its anchor.
 3. Release one endpoint.
 4. Continuously interpolate the released arc/control points toward a diagonal segment.
- This is directly implementable in vector paths, spline control meshes or implicit transition fields. The code provides `loop_to_r_morph(alpha)`.

Loop/stem to R: boundary edit plus continuous field morph

Source trick: cut/unroll the lower loop into a diagonal leg; topology and interpolation are kept explicit.



The intermediate blend is a transition field; it is not guaranteed to remain an exact signed-distance function.

Figure 4.1: The D/B/phi-to-R motif normalized as a path/field morph.

4.3 Torque axis, time and phase

TRANSLATE The source calls the stem a torque axis T and uses ΔT and $\Delta\phi$ for motion. UGTS-0 separates the overloaded symbols:

- continuous time t and time increment Δt ,
- orientation/rotation parameter θ ,
- physical torque τ only when a mechanical model is supplied,
- phase $\phi \in S^1$ and phase rate $\dot{\phi}$.

This type separation prevents a triangle, torque, time step and transformation from silently sharing one symbol.

4.4 One-bit hinge and global morph

RETAIN A bit may select closed/open, route A/B, admitted/rejected or stale/fresh according to a declared schema. **TRANSLATE** “The hinge is everywhere” becomes a global transition field or state patch evaluated at all queried points. A linear field blend is useful for visualization but intermediate fields need not be exact SDFs, and a topology change still requires an explicit gluing/transition rule.

4.5 Split oval, opposing arches and overlap lens

S4 contributes two usable constructions. First, a split oval under antisymmetric displacement becomes two curve branches around a fixed node. The source's 1:1 amplitude is a symmetric example, not universal warped-disk physics. Second, two overlapping semicircles create a measurable lens region:

$$A_{\text{lens}} = 2r^2 \cos^{-1}\left(\frac{d}{2r}\right) - \frac{d}{2} \sqrt{4r^2 - d^2}.$$

The lens can represent geometric intersection, a tolerance buffer, a blend zone or shared information $A \cap B$; its semantics must be specified by the application.

PART 5

Implicit geometry and analytic events

5.1 Implicit fields and sign convention

RETAIN A field $f : \mathbb{R}^n \rightarrow \mathbb{R}$ uses $f = 0$ as boundary, $f < 0$ as interior and $f > 0$ as exterior. Exact signed-distance status is reserved for valid primitives and distance-preserving transforms. A generic implicit field is not automatically an exact SDF.

5.2 Constructive solid geometry

The KC graphics document supplies the standard operators:

$$f_{A \cup B} = \min(f_A, f_B), \quad f_{A \cap B} = \max(f_A, f_B), \quad f_{A \setminus B} = \max(f_A, -f_B).$$

UGTS-0 implements these plus a documented smooth union and a non-distance-preserving morph field.

5.3 The zero surface becomes an event relation

This is the most important correction in the corpus. Earlier material tends to treat $f = 0$ as a graphics boundary to sample or march through. S6 instead promotes it to an event relation $R_j(q) = 0$. The system solves the crossing time and invokes transition semantics. Rendering can still sample the field, but the authoritative state change is an event.

5.4 Closed-form and numerical solvers

RETAIN For a line relation $n \cdot x = c$ and a quadratic trajectory

$$x(t) = p_0 + v_0 \Delta t + \frac{1}{2} a \Delta t^2,$$

the crossing is a quadratic root. For a circle and linear trajectory, the crossing is also quadratic.

REJECT Arbitrary SDF/trajectory pairs do not have universal exact non-iterative roots. UGTS-0 falls back to sampled bracketing and bisection for a declared generic field.

BOUNDARY / RISK

Tangencies, coincident surfaces, multiple roots, long horizons, singular log coordinates and rank changes are part of the engineering problem. Solvers must return tolerance, confidence/status and reason codes rather than pretending every event is exact.

PART 6

Local supports: cone, sphere, shell and hourglass

6.1 Cone as analytic relevance

RETAIN A cone is a local admission domain: field of view, directional influence, collision scope, memory-retrieval scope or coupler acceptance. In 2D the package uses radius, center angle, half-width and time interval. It need not be drawn.

6.2 Spherical means local radial-angular chart

S5 and S6 explicitly state that spherical does not require a global spherical world. It means a local chart around a sensor or coupler containing angular support, radial reach, orientation, uncertainty and field of view. The global substrate may remain Euclidean, graph-backed, projective or hybrid.

6.3 Nested shells

TRANSLATE Concentric shell intervals can organize multiresolution range or relevance. They are useful as a local index but are not required everywhere. A shell hierarchy is orthogonal to entity identity and does not replace a generic spatial index for unsupported scenes.

6.4 Quad hourglass and invariant axis

TRANSLATE The source's four chambers meeting at a pinch become a finite routing table. The pinch is an event locus, not a physical singularity. An invariant/eigen-axis is retained as a stable mode/direction under a transformation; eigenvalue rhetoric is not used as a magical binary switch.

PART 7

Topology as explicit gluing

7.1 Phase sheets and double vacuum

RETAIN The strongest topological/information concept is that two states may share the same coordinate yet fail to interact. The “second vacuum” is absence of coupling, not deeper emptiness. The compatibility test can include sheet, phase, orientation, branch, tags, policy and lineage.

WORKED EXAMPLE

At $t = 2$, entities A and B can both occupy $(0, 0.25)$. If A is on sheet 0 with phase 0 and B is on sheet 1 with phase π , `can_couple` returns false with `sheet_mismatch` and `phase_mismatch`. Co-location is necessary but not sufficient.

7.2 Mobius and Klein quotient maps

RETAIN A Mobius band can be implemented as a rectangle quotient $(0, y) \sim (w, -y)$, which flips orientation after one wrap. **TRANSLATE** A Klein-bottle motif becomes an explicit two-axis quotient where one wrap flips orientation/sheet and the other closes periodically. It is a software coordinate/gluing map, not a claim that matter passes through itself physically.

7.3 Topological portals

A portal is specified by:

- entry event surface and admission support,
- exit coordinate transform,
- orientation and sheet map,
- phase/branch policy,
- lineage event and uncertainty.

This is sufficient for non-orientable game spaces without requiring a physically embedded Klein bottle.

Topology as explicit state gluing and routing

Non-orientable motifs become quotient maps; double vacuum becomes compatibility-separated co-location.

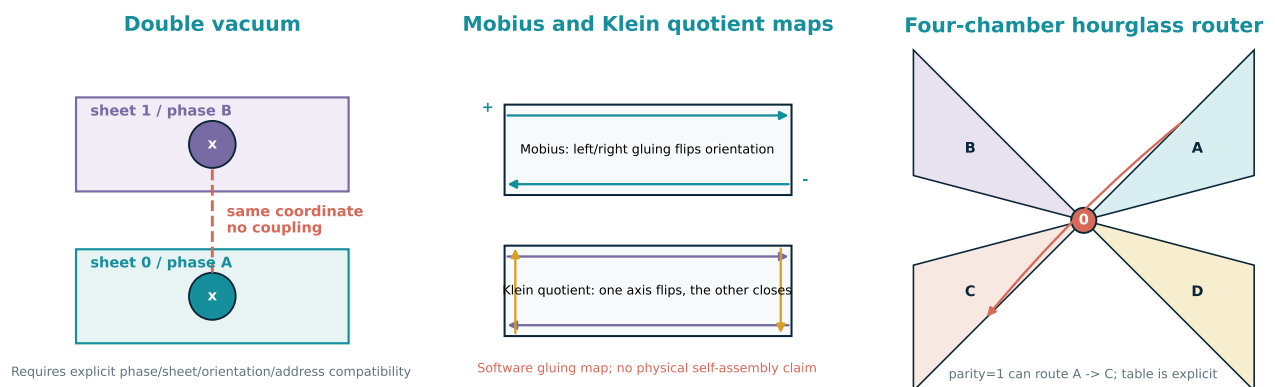


Figure 7.1: Double vacuum, quotient gluing and hourglass routing after normalization.

7.4 Inside/outside is schema-bound

The sign of an implicit field, the parity bit and the orientation bit are separate typed values. “Inside” cannot be inferred from parity without a schema. This mirrors the corpus’s COBOL packed-decimal lesson: raw bits are meaningless without layout metadata.

PART 8

Information architecture formalism

8.1 Typed state manifold

A compact reference state is

$$q_e(t) = (x_e(t), t, \phi_e(t), s_e(t), o_e(t), a_e, b_e(t), u_e(t)),$$

where x is position, ϕ phase, s sheet, $o \in \{-1, +1\}$ orientation, a lineage address, b branch and u uncertainty. A corresponding abstract space is

$$Q = P^n \times \mathbb{R}_{time} \times S_{phase}^1 \times S_{sheet} \times O_{orientation} \times A_{address} \times B_{branch} \times U.$$

8.2 World definition

The world is not only geometry:

$$W = (G, R, T, I, L),$$

with finite grammar G , relation family R , transition family T , invariants I , and external event log L .

8.3 Support, compatibility, event and transition

For relation j and support α :

$$R_j(q; \theta_j) = 0, \quad C_\alpha(q, t) \leq 0, \quad \chi(e, j, t) \in \{0, 1\}.$$

The next event is

$$t^* = \min\{t > t_0 : R_j(q_e(t)) = 0, C_\alpha(q_e(t), t) \leq 0, \chi(e, j, t) = 1\}.$$

The transition produces a patch and event record:

$$q_e(t^{*+}) = T_j(q_e(t^{*-}), \text{context}).$$

State and event formalism

The mature corpus becomes a typed compatibility-gated event calculus

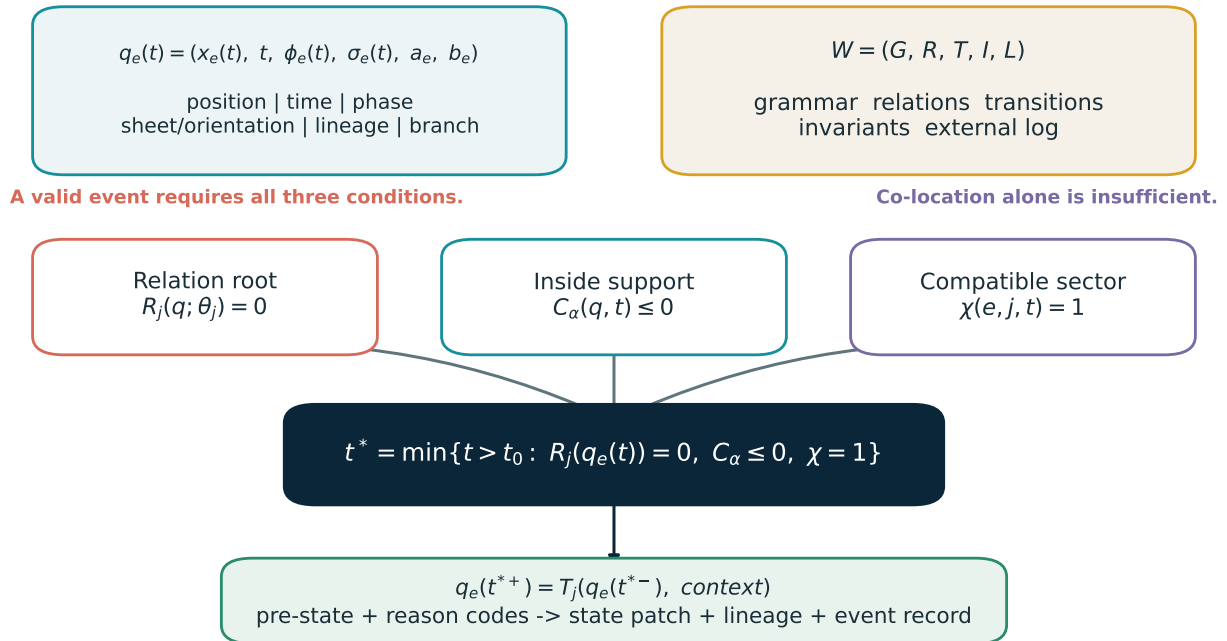


Figure 8.1: The formal event condition requires relation, support and compatibility.

8.4 Reason codes and confidence

A boolean is insufficient for engineering. Compatibility returns reason codes such as sheet mismatch, phase mismatch, missing tag or policy rejection. An event record contains pre/post state, solver family, crossing direction, relation residual, confidence, lineage and metadata.

PART 9

Finite grammar, identity and memory

9.1 Finite grammar

RETAIN L-system and shape-grammar language is useful only after depth and symbol budgets are explicit. The implementation refuses expansion beyond `max_depth` or `max_symbols`; unknown variables are inert rather than magical.

9.2 Lineage address

Coordinates are not identity. An entity keeps a stable ID plus transition ancestry. Portals, branch changes, splits and merges must append lineage records or create explicit parent/merge references. This allows identity reconstruction even after co-location or coordinate reuse.

9.3 External event log

Closed deterministic dynamics may be recomputed from parameters, but exogenous novelty cannot. The irreducible log records inputs, branch choices, calibration changes, policy decisions and uncertain solver outcomes. Therefore the substrate can reduce frame storage in restricted cases but cannot be zero-memory.

9.4 Schema-bound packed data

S2/S8/S9 discuss COBOL COMP-3 and copybooks. The useful lesson is architectural: a byte stream without layout is not self-describing. UGTS-0 versions world schemas, event schemas, LUT dimensions, sign conventions and parity meanings.

PART 10

Query-first runtime and complexity

10.1 Minimum API

The six minimum queries are:

1. `state_at(entity,t)`
2. `next_event(entity,t0,t1)`
3. `events_in_support(support,t0,t1)`
4. `can_couple(a,b,t)`
5. transition application / event processing
6. `reconstruct_identity(entity)`

10.2 Precise meaning of horizon skipping

For a fixed closed expression, `state_at(t)` can be independent of the number of skipped simulation frames. This is not universal $O(1)$: expression size, grammar depth, branch history and transcendental evaluation still cost resources. `next_event` is dominated by candidate filtering, root solving and branch resolution.

10.3 Success criterion

The architecture wins only if

$$\text{cost}(\text{support} + \text{compatibility} + \text{event solve}) < \text{cost}(\text{explicit materialization avoided})$$

for the target relation family and error boundary. Poor compatibility predicates turn cone queries into broad scans; dense events or grammar growth can erase any advantage.

PART 11

Game-technology integration

11.1 Hybrid engine structure

A practical game engine does not need to abandon conventional systems. The authoritative UGTS state can coexist with meshes, rigid bodies, audio, animation, editor tooling and rasterization.

Game-technology integration

A hybrid engine keeps UGTS-0 authoritative while conventional rendering, audio and tooling remain available

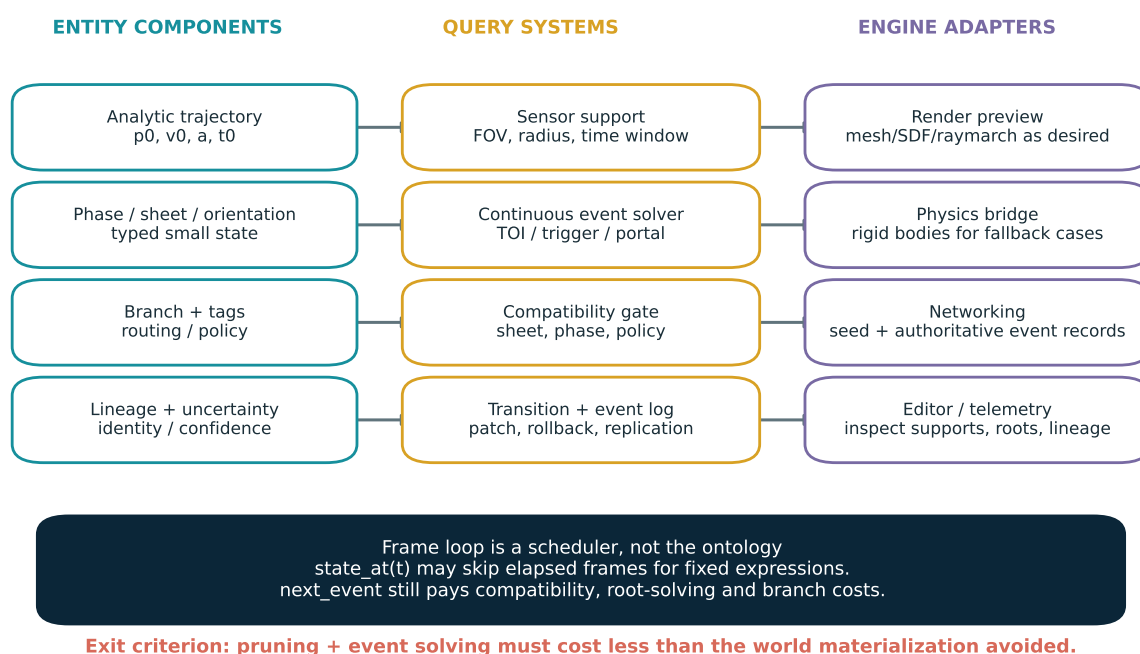


Figure 11.1: Component/system mapping for Unity, Godot or an ECS runtime.

11.2 Components

Analytic motion component

Coefficients for linear or quadratic trajectory plus start time.

Phase/sheet/orientation component

Small typed state, never collapsed into one unnamed bit.

Support component

Local FOV, radial range and time window.

Relation component

Line, circle or generic implicit event surface.

Transition component

Sheet/orientation/phase/branch/tag patch with lineage label.

Event-log component

Authoritative event records for rollback and replay.

11.3 Continuous collision and portals

For supported relations, time of impact is solved analytically. Unsupported cases can fall back to a conventional physics engine or numerical method. A portal is just an event surface plus explicit exit map and state patch, making Mobius/Klein/hourglass effects deterministic and inspectable.

11.4 Networking

Seed/parameter replication plus authoritative event records can reduce frame replication for stable closed dynamics. Cross-platform floating-point differences, nondeterministic branches and exogenous inputs still require synchronization, schema versions and correction snapshots.

11.5 Adapters supplied

The package includes minimal Unity C# and Godot GDScript adapters. They demonstrate `state_at` and a line-crossing query without claiming full engine integration.

PART 12

Optional graphics projection

12.1 Log-polar chart

RETAIN The local map is

$$\rho = \ln(r/r_0), \quad \theta = \text{atan2}(y, x).$$

Scaling in real space becomes translation in ρ . The origin remains singular, so UGTS-0 uses an explicit epsilon/core flag; the logarithm does not remove the singularity.

12.2 One-bit log-polar LUT

A compact bitset marks active (ρ, θ) sectors. It is a mask for admission or projection work, not complete world state. LUT dimensions, range, origin convention and bit meaning are schema fields.

12.3 Phasor edge accumulation

TRANSLATE The source calls its subpixel vectors “Feynman vectors.” UGTS-0 normalizes this to ordinary complex phasors whose phase follows local field-gradient orientation and whose magnitude follows coverage. This can be used for oriented supersampling; no quantum path-integral claim is made.

12.4 One-bit jitter and temporal modulation

RETAIN A static projection can use deterministic stochastic thresholds. **TRANSLATE** A display can use pulse-density/sigma-delta bitstreams to approximate intensity over time. These are optional output encodings and require hardware calibration.

12.5 Chromatic and prepress transforms

A real-space magnification $r \mapsto sr$ becomes $\rho \mapsto \rho + \ln s$, so channel-specific chromatic scale can be expressed as a log-radius offset. It is cheap, not free. The prepress adapter maps channels to CMYK, uses log-radius-dependent screening and must compensate dot gain and moire risk.

Optional graphics adapter

The authoritative substrate emits geometry/state; a replaceable adapter projects it to pixels or print

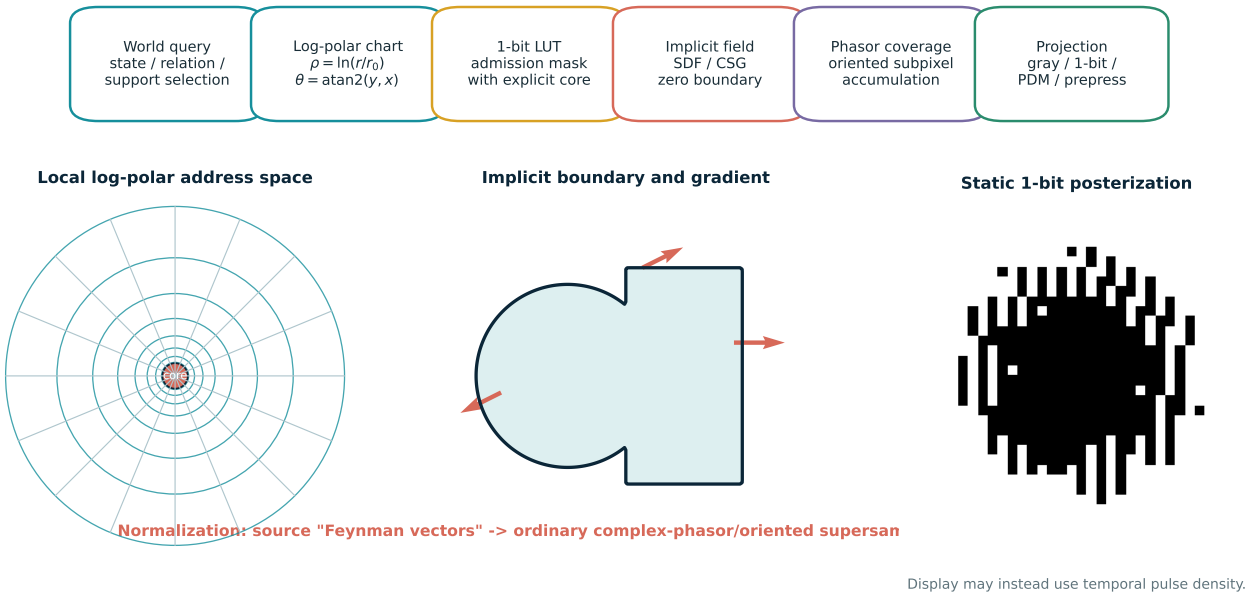


Figure 12.1: The KC graphics material becomes a replaceable projection pipeline.

PART 13

Bounded Compatibility Event hardware endpoint

13.1 Canonical signal path

S5 gives the best-defined physical translation:

local support → liquid-controlled coupling → compatible mode → measured guard → verified output.

A local radial-angular optical field is focused or phase-shaped by a liquid region, coupled into selected guided modes, filtered by wavelength/polarization/mode/phase compatibility, and declared an event only at a measured guard crossing.

13.2 Type safety across physics

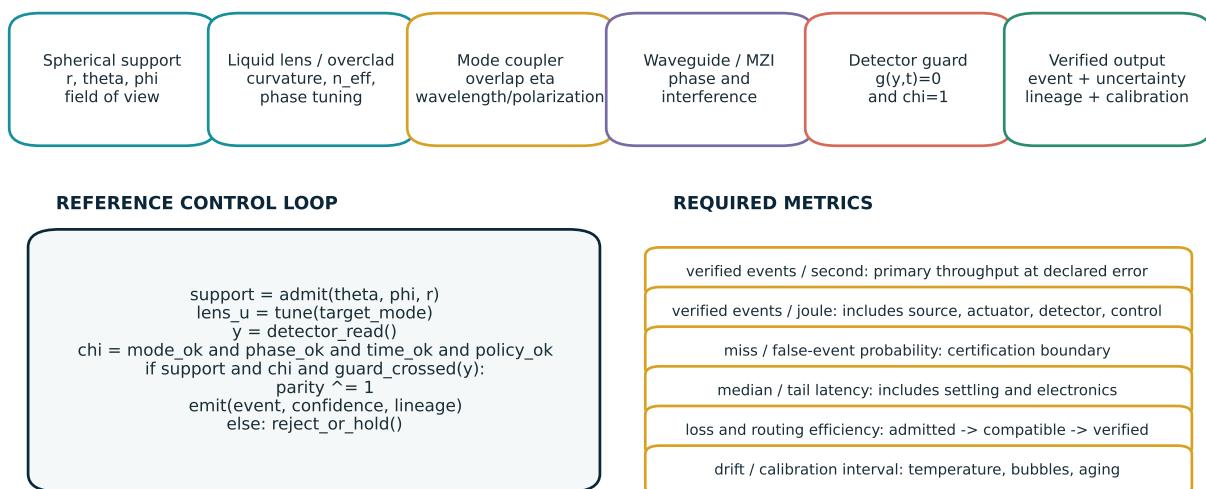
Maxwell governs the optical field; Young-Laplace/fluid mechanics governs the meniscus; coupled-mode theory governs guided amplitudes; digital event logic decides when a result is declared. A Klein motif may describe routing only after ports, orientation map and transfer matrix are specified.

13.3 Narrow parity role

The parity bit can mark route, admission, freshness or latch state. Detector amplitudes, thresholds, confidence, uncertainty, calibration and lineage remain separate. The reference controller detects a rising/falling threshold crossing only after support and compatibility are true.

Bounded Compatibility Event hardware endpoint

A disciplined optofluidic translation: measured guard crossings, not topological magic



One-bit parity is only a route/latch flag. Optical amplitude, threshold, uncertainty and lineage remain separate.

Prototype is killed if calibration, loss, actuation bandwidth or energy erases the measured advantage.

Figure 13.1: B.C.E. signal path, control loop and required metrics.

13.4 Hollowlens-0

The source proposes one scanned input, one tunable liquid coupler, a 2×2 interferometric waveguide, balanced detectors and an FPGA/MCU sidecar. Success requires preregistered wavelength, power, support, compatible modes, thresholds, uncertainty, energy boundary, calibration interval and matched electronic baseline.

13.5 Kill criteria

The prototype is stopped or simplified if optical loss, liquid instability, slow actuation, mode crosstalk, calibration drift, event ambiguity, packaging failure or sidecar dominance removes the energy/accuracy advantage.

PART 14

Reference implementation

14.1 Package structure

Listing 14.1: Delivered package layout

1	README.md	technical entry point
2	specs/	source map, 69 concepts, claims, schemas
3	src/ugts/	dependency-free Python reference runtime
4	tests/test_ugts.py	executable verification
5	examples/	world, topology, graphics, fractal, B.C.E. demos
6	shaders/ugts_preview.frag	optional GLSL projection
7	adapters/unity/	minimal C# game adapter
8	adapters/godot/	minimal GDScript game adapter
9	diagrams/	report figures
10	source_notes/	per-PDF extraction notes

14.2 Core modules

Module	Responsibility
state.py	Typed entity state, state patches, lineage and event records.
trajectory.py	Closed-form linear and quadratic motion.
support.py	Local radial-angular and time admission.
compatibility.py	Sheet, orientation, phase, branch, tags and lineage gating.
events.py	Analytic line/circle roots and bracketed generic-field roots.
transition.py	Pure schema-bound state patches.
world.py	Query API, transition processing, double-vacuum coupling and event log.
geometry.py	Implicit primitives, CSG, morph, gradient and overlap lens.
topology.py	Mobius/Klein quotient maps, portals and hourglass routing.
logpolar.py	Explicit-core transform and compact bit LUT.
numeric.py	Radix, Hamming, Pascal parity and golden-angle samples.
glyphs.py	Loop/stem and loop-to-R transition fields.
grammar.py	Finite budgeted grammar and small shape compiler.
render.py	Optional CPU projection, posterization and PDM.
bce.py	Measured compatibility/guard controller.
io.py	JSON world loader and event-log serialization.

14.3 World query example

Listing 14.2: Minimal query-first use

```
1 from ugts import load_world
2
3 world = load_world("specs/example_world.json")
4 state = world.state_at("traveler_A", 1.25)
5 candidate = world.next_event("traveler_A", 0.0, 5.0)
6 record = world.process_next_event("traveler_A", 0.0, 5.0)
7
```

```
8 coupling = world.can_couple(  
9     "traveler_A", "co_located_B", 2.0,  
10     max_distance=1e-9,  
11     phase_tolerance=0.2,  
12     require_same_sheet=True,  
13     require_same_orientation=True,  
14 )
```

14.4 Event solver behavior

Line surfaces support linear and quadratic trajectories. Circle surfaces currently support analytic roots for linear trajectories. Generic fields use a sampled sign-change search followed by bisection. The solver excludes the current instant by an epsilon, applies support and compatibility, then sorts candidate events by time and rule ID.

14.5 Topological map behavior

Listing 14.3: Explicit quotient maps

```
1 from ughts import MobiusBand, KleinBottleQuotient, Vec2  
2  
3 mobius = MobiusBand(width=4.0, height=2.0)  
4 print(mobius.map(Vec2(4.5, 0.25)))  
5 # x wraps to 0.5; y and orientation flip  
6  
7 klein = KleinBottleQuotient(width=4.0, height=2.0)  
8 print(klein.map(Vec2(4.5, 0.25)))  
9 # orientation flips and sheet toggles
```

PART 15

Verification and test results

15.1 Executable tests

The package includes seventeen tests across numeric seeds, implicit geometry, event solving, transitions, double-vacuum coupling, quotient maps, log-polar roundtrip, bit-LUT storage, rendering, grammar budgets, B.C.E. and JSON I/O.

Listing 15.1: Test run captured during package construction

```
1 test_guard_crossing_and_parity (test Ugts.BCETests.test_guard_crossing_and_parity) ... ok
2 test_circle_event (test Ugts.EventWorldTests.test_circle_event) ... ok
3 test_double_vacuum_same_coordinate_no_coupling (test Ugts.EventWorldTests.
  test_double_vacuum_same_coordinate_no_coupling) ... ok
4 test_next_event_and_transition (test Ugts.EventWorldTests.test_next_event_and_transition) ... ok
5 test_csg_signs (test Ugts.GeometryTests.test_csg_signs) ... ok
6 test_glyph_morph_is_queryable (test Ugts.GeometryTests.test_glyph_morph_is_queryable) ... ok
7 test_overlap_lens_area (test Ugts.GeometryTests.test_overlap_lens_area) ... ok
8 test_finite_expansion (test Ugts.GrammarTests.test_finite_expansion) ... ok
9 test_load_example_and_write_log (test Ugts.IOTests.test_load_example_and_write_log) ... ok
10 test_lut_storage_and_render (test Ugts.LogPolarAndRenderTests.test_lut_storage_and_render) ... ok
11 test_roundtrip_and_core (test Ugts.LogPolarAndRenderTests.test_roundtrip_and_core) ... ok
12 test_sigma_delta (test Ugts.LogPolarAndRenderTests.test_sigma_delta) ... ok
13 test_active_bits_and_pascal (test Ugts.NumericTests.test_active_bits_and_pascal) ... ok
14 test_radix_and_indexing (test Ugts.NumericTests.test_radix_and_indexing) ... ok
15 test_klein_sheet_flip (test Ugts.TopologyTests.test_klein_sheet_flip) ... ok
16 test_mobius_orientation_flip (test Ugts.TopologyTests.test_mobius_orientation_flip) ... ok
17 test_portal_and_hourglass (test Ugts.TopologyTests.test_portal_and_hourglass) ... ok
18
19 -----
20 Ran 17 tests in 0.008s
21
22 OK
```

15.2 What the tests establish

The tests establish internal consistency of the bounded implementation. They do not validate the corpus's external performance, physical hardware throughput or universal claims. In particular, they demonstrate that:

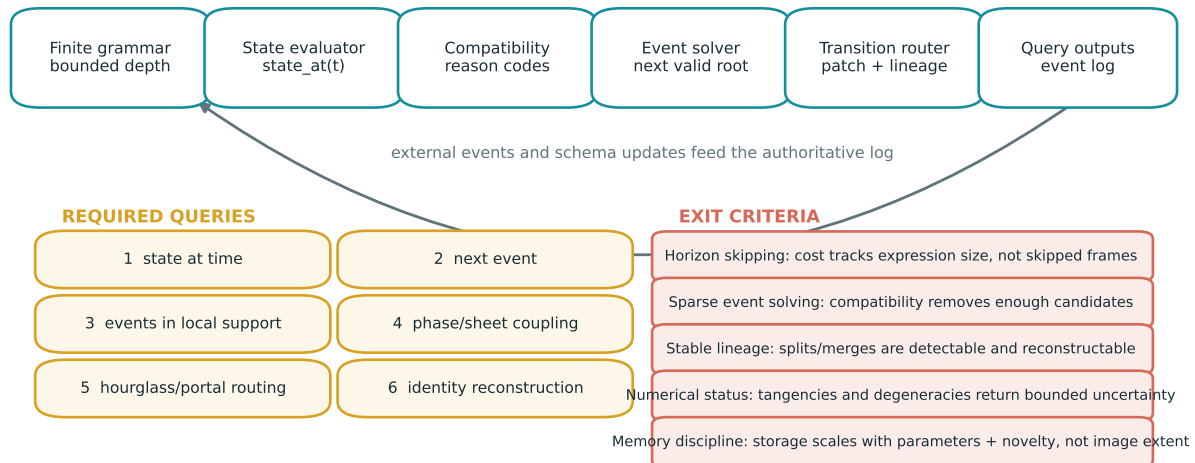
- the traveler crosses $x = 0$ at $t = 2$ and receives the expected sheet/orientation/branch patch;
- co-located incompatible states do not couple;
- Mobius and Klein quotient maps apply declared orientation/sheet changes;
- Pascal row 7 is all odd while row 8 is not;
- log-polar conversion requires an explicit core chart;
- the one-bit LUT uses the expected compact storage;
- the B.C.E. bit toggles only after a valid measured crossing.

PART 16

Prototype program and falsification

Equation World Zero: minimum falsifiable prototype

Six queries and explicit exit criteria test the core without turning it into a renderer benchmark



Failure is useful: event explosion, grammar growth, poor conditioning or weak compatibility directly falsifies the claimed advantage.

Figure 16.1: Equation World Zero is a bounded falsifiable experiment.

16.1 Required experiments

Horizon skipping

Measure whether `state_at(t)` depends on expression size rather than the number of skipped frames.

Co-located phase sheets

Verify that same-position states remain uncoupled until explicit compatibility is met.

Event routing

Verify ordering, transition invariants and reason codes at line/circle/hourglass events.

Grammar-depth stress

Measure expression growth and normalization under repeated production.

Identity split/merge

Add explicit parent/merge records and detect lineage collisions.

Minimal agent integration

Let perception and action use the same support/event language without claiming general intelligence.

16.2 Failure modes

The core fails usefully if event density explodes, compatibility filtering is weak, grammar composition escapes the normalized relation family, numerical conditioning is poor, lineage cannot survive split/merge, or exogenous novelty dominates storage.

16.3 Implementation roadmap

1. Freeze schema version 1.0 and the restricted relation families.
2. Add interval-valued root/status results for tangencies and uncertain coefficients.
3. Add explicit split/merge lineage records and deterministic serialization hashes.
4. Benchmark against fixed-step and conventional CCD baselines at equal error.
5. Integrate one engine adapter with editor visualization but keep authority in the query core.
6. Only after software benchmarks, build Hollowlens-0 with a preregistered electronic baseline.

PART 17

Claims ledger and final decision

17.1 Claims not carried into UGTS-0

The following are not requirements or conclusions: universal constant-time worlds; zero memory, power, heat or latency; exact non-iterative intersections for arbitrary fields; complete state in one bit; physical Klein-bottle self-assembly; topology that eliminates all broad phase; golden-ratio zero-cost anti-aliasing; replacement of all renderers/PDE solvers; or replacement of general AI.

Table 17.1: Claims ledger: corrections and rejected totalizing claims.

ID	Source claim	Disposition	Technical reason
CL01	Universal $O(1)$ world/event solving	REJECT	Only fixed closed expressions and restricted fixed-degree root problems may be horizon-independent; candidate count and solver cost remain.
CL02	Zero memory / under-32MB universal worlds	REJECT	Parameters can compress some closed dynamics, but entities, branches, exogenous events, calibration and uncertainty consume memory.
CL03	No latency, heat or power	REJECT	All computation, actuation, detection and display/hardware paths have latency and energy costs.
CL04	One bit contains the complete state	REJECT	One bit is only a schema-bound flag; numeric state, uncertainty, lineage and parameters remain separate.
CL05	Golden ratio eliminates aliasing at zero cost	DEMOTE	Golden-angle schedules can reduce regularity but require sampling, reconstruction and error analysis.
CL06	Feynman vectors are a standard quantum rasterizer	TRANSLATE	Use the phrase only as inspiration; the implementation is complex phasor/oriented supersampling.
CL07	Klein bottle physically self-assembles matter	REJECT	Use quotient/gluing maps in software; no physical self-assembly claim is supported.
CL08	Möbius strip is simply half a Klein bottle	CORRECT	Certain cuts/decompositions relate them, but the statement is not a general unique identity.
CL09	Rows at powers of two in Pascal are all ones/odd	CORRECT	With zero-based row indexing, rows 2^k-1 are all odd; Pascal parity still yields the Sierpinski pattern.
CL10	$C(5,3)=10$ proves the decimal threshold relation	DEMOTE	It is a valid combination count and an interesting coincidence, not a causal structural proof.
CL11	Log mapping removes the origin singularity	CORRECT	Log mapping relocates/rescales the singular behavior; an epsilon/core chart is still required.

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Table 17.1 continued

ID	Source claim	Disposition	Technical reason
CL12	Analytic SDF intersections are always exact and non-iterative	REJECT	Only selected field/trajectory pairs have closed-form roots; generic cases need numerical/certified methods.
CL13	All BVH/broad-phase structures become unnecessary	REJECT	Strong support/compatibility filters may replace or complement broad phase in restricted domains; generic scenes still need indexing.
CL14	The topology solves the Von Neumann bottleneck	REJECT	Data locality may improve, but instruction/data movement and memory hierarchy remain physical constraints.
CL15	World geometry directly replaces general AI/LLMs	REJECT	A shared query substrate can host agents, but it does not replace model weights, learning, semantics or inference complexity.
CL16	Biology, dreams, economies and wireless auras instantiate the same exact mechanism	REJECT	These are narrative analogies in the source bundle, not part of the technical unification.
CL17	Chromatic correction is zero-cost	CORRECT	Log-space offsets are cheap, but channel evaluation, filtering, resampling and calibration still cost resources.
CL18	Rasterization and raymarching are forbidden	CORRECT	They are optional downstream projection tools; they do not define the authoritative substrate.

DECISION

GO for formalization and restricted Equation World Zero. Start with algebra of state, support, compatibility, event, transition, identity and memory. Use rasterization, raymarching, game engines and waveguides as downstream adapters or benchmarks. Do not begin by recreating a conventional scene and then mistake the projection for the substrate.

PART A

Complete extracted concept inventory

The following table is the comprehensive extraction delivered with the package. The machine-readable version is `specs/concept_inventory.csv`.

Table A.1: Complete extracted concept inventory. “Source expression” preserves the corpus framing; “normalized operator” is the technical implementation used by UGTS-0.

ID	Category / name	Source expression	Normalized operator and qualification	Status	Source trace
N01	Radix carry threshold Numeric/fractal	Digit-count changes at powers of the radix.	Use radix powers as scale/change boundaries and compact level indices. Exact for positional number systems.	RETAIN	S1, pp. 1-4
N02	Zero-based offset semantics Numeric/fractal	An index of 35 denotes the 36th stored position.	Keep value, offset and ordinal position as distinct typed quantities. Prevents off-by-one errors; not altered arithmetic.	RETAIN	S4, pp. 1-3
N03	Active-bit/Hamming-weight filter Numeric/fractal	19 = 10011 has three active bits; zeros are treated as inactive.	Use popcount as a sparse feature count or seed descriptor. The phonetic syllable link is a mnemonic, not a general encoding law.	TRANSLATE	S1, pp. 3-5
N04	Three-point triangle seed Numeric/fractal	Three active positions are treated as triangle anchors.	Map three non-collinear active samples to a simplex/triangle primitive. Any three non-collinear points form a triangle; the mapping is chosen.	TRANSLATE	S1, pp. 6-7
N05	Pascal parity to Sierpinski Numeric/fractal	Odd/even entries in Pascal triangle form a Sierpinski pattern.	Use binomial parity as a deterministic fractal/test-pattern generator. Technical correction: rows 2^k-1, not generally 2^k, are all odd when indexed from zero.	RETAIN	S1, pp. 6-9
N06	Golden-angle/Fibonacci sampling Numeric/fractal	Phi/Vogel spacing is proposed to avoid visible repetition.	Optional quasi-uniform angular sample schedule or dither seed. Useful heuristic; it does not guarantee zero aliasing or zero cost.	TRANSLATE	S1, pp. 13-15
N07	Lemniscate/strange-loop recursion Numeric/fractal	Infinity and Escher loops symbolize self-generation.	Represent recursion with explicit finite grammar depth and cycle detection. Keep as a recursion motif, not a proof of self-creating physics.	DEMOTE	S1, pp. 18-23
G01	Loop-plus-stem decomposition Glyph/kinematic	Chosen phi/D/B glyphs are decomposed into loop(s) plus a stem.	Represent glyph geometry as a graph of curve segments and junctions. Glyph form varies by font; the decomposition is a design convention.	TRANSLATE	S9,S2,S8, pp. 1-2
G02	Cut-unroll D/B to R Glyph/kinematic	Break the lower loop and straighten the released arc into a diagonal leg.	Boundary-graph edit plus continuous curve morph from closed arc to open segment. Implementable as path topology and control-point interpolation.	RETAIN	S9,S2,S8, pp. 1-4
G03	Torque axis T Glyph/kinematic	The stem is an axis; rotation shears/unreels the lower loop.	Use a rotation/warp parameter around a declared pivot axis. Torque language is a kinematic metaphor unless a physical model is supplied.	TRANSLATE	S9,S2,S8, pp. 1-4
G04	Delta-time and phase modulation Glyph/kinematic	Delta T and delta phi drive movement and phase change.	Store continuous time and phase as separate state coordinates and derivatives. Do not overload time, torque and triangle into one untyped symbol.	RETAIN	S1,S7,S9, pp. 10-15;14-18;2-5
G05	One-bit parity hinge Glyph/kinematic	A bit toggles closed/open or route A/B.	Use a schema-bound route/parity flag that selects transition behavior. The bit is not the complete state.	RETAIN	S5,S6,S7,S9, pp. 2-8;5-7;14-25;3-8
G06	Global hinge field Glyph/kinematic	The hinge is applied everywhere through an SDF sign/state change.	Model a global morph or transition field evaluated at every query point. A morph parameter changes the implicit field; topology changes require explicit handling.	TRANSLATE	S9,S2,S8, pp. 7-8
G07	Chirality/orientation flip Glyph/kinematic	R/S or mirrored states are associated with sign/orientation reversal.	Track an orientation bit and apply a gluing/transition map that may flip it. Orientation is distinct from sheet and phase.	RETAIN	S1,S7,S9, pp. 21-29;18-25;4-8
G08	Split oval to opposing arches Glyph/kinematic	A zero-centered oval is split and torque opens it into positive/negative wave arches.	Use symmetric curve branches around a fixed node for waveform or deformation profiles. The exact 1:1 amplitude condition is a chosen symmetric case, not universal warped-disk physics.	TRANSLATE	S4, pp. 10-16

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Table A.1 continued

ID	Category / name	Source expression	Normalized operator and qualification	Status	Source trace
G09	Overlap lens/shared domain Glyph/kinematic	Slightly overlapping semicircles form an interference/tolerance/shared-information zone.	Use intersection $A \cap B$, lens area, or blend region as explicit shared domain. Interpretation depends on geometry, mechanics or information semantics.	RETAIN	S4, pp. 16-19
I01	Signed distance/implicit field Implicit geometry	Geometry is represented by $f(p)$, with $f=0$ as the boundary.	Use implicit scalar fields; reserve exact-distance claims for valid primitives/transforms. A generic implicit field need not be an exact SDF.	RETAIN	S1,S3,S6,S7,S9, pp. 10-30;1-10;5-10;14-31;7-8
I02	SDF CSG operators Implicit geometry	Union= $\min$, intersection= $\max$, subtraction= $\max(a,-b)$.	Provide composable implicit-field operators with documented sign convention. Smooth variants may be added as optional adapters.	RETAIN	S3, pp. 1-3
I03	Zero surface as event relation Implicit geometry	The SDF-zero locus becomes a transition surface, not a place to march through.	Treat $R_j(q)=0$ as an event guard and solve crossing time. This is the central correction from rendering to event semantics.	RETAIN	S6,S5,S7, pp. 5-7;3-7;14-31
I04	Analytic kinematic sweep Implicit geometry	A cone/trajectory is solved against an implicit boundary for exact event time.	Use closed-form roots for restricted trajectory/surface pairs; certified numerical roots otherwise. No universal non-iterative solver exists.	RETAIN	S6,S7, pp. 7-12;14-31
I05	Cone as support/domain Implicit geometry	Cones are directional influence, field-of-view or admissibility domains.	Use analytic radial-angular support predicates for pruning/query scope. Not necessarily a rendered cone.	RETAIN	S5,S6,S7, pp. 3-6;5-10;12-25
I06	Spherical/local radial-angular support Implicit geometry	Spherical means a local chart around a sensor/coupler, not a global world remesh.	Represent local support by radius, angle, orientation, uncertainty and time window. Core type-safety rule.	RETAIN	S5,S6, pp. 3-6;1-10
I07	Nested shells Implicit geometry	Shells organize radial reach, scale and relevance.	Use concentric support bands or multiresolution radial intervals. Useful local indexing structure, not required globally.	TRANSLATE	S6,S7, pp. 1-7;12-25
I08	Hourglass/quad routing Implicit geometry	Four conical chambers meet at a pinch/event locus.	Implement a finite routing partition keyed by sign/sheet/orientation at a transition surface. The pinch is an event/router, not a literal singularity that destroys data.	TRANSLATE	S6,S7, pp. 5-7;23-25
I09	Invariant/eigen-axis Implicit geometry	A central axis or mode persists through transformations.	Declare invariants and preserved quantities for transitions. Eigenlanguage is only literal when a linear operator is actually defined.	RETAIN	S1,S6,S7, pp. 15-20;5-12;18-25
I10	Projective/homogeneous chart Implicit geometry	Null-cone and projective language compactifies direction/scale.	Optional homogeneous coordinates for intersections and points at infinity. Use only where it improves equations and conditioning.	TRANSLATE	S6,S7, pp. 5-7;18-25
C01	Log-polar transform Coordinates/sampling	$\rho=\ln(r)$, $\theta=\text{atan2}(y,x)$.	Use a local coordinate chart that turns radial scaling into translation in ρ . Handle $r=0$ with an epsilon/core cell.	RETAIN	S1,S3,S7,S9, pp. 10-15;1-10;12-31;5-8
C02	One-bit log-polar LUT Coordinates/sampling	A bitmask activates radial-angular sectors.	Use as a coarse admission/cache mask, never as the whole world state. Schema and quantization parameters are mandatory.	RETAIN	S1,S3,S7,S9, pp. 10-15;1-10;12-31;5-8
C03	Phasor/Feynman-style edge accumulation Coordinates/sampling	Subpixel samples carry magnitude/phase and are accumulated.	Normalize to complex phasor or oriented coverage accumulation for antialiasing. Not a literal Feynman path integral unless a physical path-integral model is defined.	TRANSLATE	S3, pp. 1-8
C04	One-bit stochastic jitter Coordinates/sampling	Binary noise shifts samples/thresholds around an edge.	Use deterministic blue-noise/hash dithering with reproducible seeds. Jitter trades structured error for noise; it does not remove error.	RETAIN	S1,S3, pp. 10-15;7-10
C05	Temporal pulse-density modulation Coordinates/sampling	Brightness/coverage is encoded as density of one-bit pulses.	Optional display-output adapter using PDM or delta-sigma modulation. Panel hardware and refresh limits remain.	TRANSLATE	S3, pp. 4-6

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Table A.1 continued

ID	Category / name	Source expression	Normalized operator and qualification	Status	Source trace
C06	Chromatic log-radius offset Coordinates/sampling	Channel scaling becomes additive rho offsets.	Apply channel-specific log-radius shifts before field evaluation, then resample correctly. Cheap coordinate arithmetic is not zero-cost end-to-end correction.	RETAIN	S3, pp. 7-8
C07	Log-polar preprocess screening Coordinates/sampling	Spatial one-bit dots scale with rho and orient with theta.	Optional print adapter with calibrated halftone, screen angles and dot-gain compensation. Requires press/paper calibration.	TRANSLATE	S3, pp. 8-10
T01	Phase sheets Topology	Multiple states occupy layered sheets over the same base coordinate.	Represent state as base position plus sheet, phase, orientation and address. Layer identity must be explicit.	RETAIN	S6,S7,S9, pp. 5-10;14-25;5-8
T02	Double vacuum Topology	Co-located states do not interact when phase/sheet/address are incompatible.	Compatibility-gated coupling: same x does not imply same sector. One of the strongest corpus concepts.	RETAIN	S5,S6,S7, pp. 2-7;5-10;14-25
T03	Mobius gluing Topology	A boundary identification reverses orientation after one loop.	Implement quotient-map boundary wrapping with orientation flip. Use a chart/gluing abstraction rather than literal 4D geometry.	RETAIN	S7,S9, pp. 7-12;5-8
T04	Klein-bottle gluing Topology	A closed non-orientable quotient is used for looping routes.	Implement two rectangle-edge identifications, one orientation reversing. Hardware realization remains frontier; 3D pictures self-intersect.	TRANSLATE	S5,S7,S9, pp. 2,5;7-25;5-10
T05	Topological portal/gluing map Topology	Crossing a boundary re-enters another chart/sheet, possibly transformed.	Use explicit port, orientation map, coordinate transform and transfer metadata. This makes topology technically testable.	RETAIN	S5,S6,S7, pp. 2-5;5-12;14-25
T06	Inside/outside sign with schema Topology	Binary sign denotes interior/exterior or route state.	Bind every bit to a declared schema/version and keep uncertainty separate. Raw bits without layout/schema are ambiguous.	RETAIN	S3,S5,S7,S9, pp. 1-10;2-7;14-31;3-8
T07	Strange loop/self-reference Topology	Escher/Klein motifs suggest systems feeding back into themselves.	Use explicit cycles, fixed points and termination/cycle policies in the grammar/event graph. Narrative inspiration only unless formally specified.	DEMOTE	S1,S7,S9, pp. 18-30;7-25;5-10
A01	Finite grammar G Information architecture	A finite rule set generates relations/structures.	Use bounded-depth typed productions that compile to supported relation primitives. Enforce symbol and expression caps.	RETAIN	S6,S7,S9, pp. 1-12;26-31;5-8
A02	State manifold Q Information architecture	State includes position, time, phase, sheet, address and branch.	Use a typed immutable state record; optional orientation and uncertainty extensions. Core data model.	RETAIN	S6, pp. 5-7
A03	Relation family $R_j(q)=0$ Information architecture	World geometry/constraints are equations rather than frame snapshots.	Store typed relations with parameters, domains and solver capability metadata. Restrict to relation families with known solvers for the prototype.	RETAIN	S6, pp. 5-12
A04	Support predicate $C \leq 0$ Information architecture	Only relations inside a declared local support are candidates.	Separate admission/support from geometry and from compatibility. Supports analytic pruning.	RETAIN	S5,S6, pp. 3-7;5-12
A05	Compatibility predicate chi Information architecture	Co-location is insufficient; phase/sheet/-mode/time/policy must match.	Compose physical and digital compatibility predicates. Return reason codes, not only a bit.	RETAIN	S5,S6, pp. 3-7;5-12
A06	Event time t^* Information architecture	The next event is the earliest supported compatible relation root.	Compute min valid root after t_0 with tolerance and confidence. Multiple roots and tangencies need policy.	RETAIN	S5,S6, pp. 3-7;5-12
A07	Transition rule T_j Information architecture	Crossing an event surface updates sheet/phase/branch/invariants.	Use pure transition functions producing state patches and event records. Preserve pre/post states and provenance.	RETAIN	S5,S6, pp. 5-7;5-12

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Table A.1 continued

ID	Category / name	Source expression	Normalized operator and qualification	Status	Source trace
A08	Lineage address Information architecture	Identity persists by generative address and invariant history.	Use stable entity IDs plus parent/merge ancestry and transition lineage. Coordinates are not identity.	RETAIN	S5,S6,S7, pp. 3-12;1-12;26-35
A09	Irreducible external event log Information architecture	Novel exogenous events must be recorded even when closed dynamics are recomputable.	Event-source only novelty, transitions, confidence and calibration changes. No zero-memory claim.	RETAIN	S5,S6, pp. 7-11;1-12
A10	Query-first API Information architecture	Ask state_at(t), next_event, events_in_support, coupling and identity.	Expose direct query methods independent of a frame loop. Projection is downstream.	RETAIN	S5,S6, pp. 3-12;1-12
A11	Projection adapter Information architecture	Images are optional outputs, not the authoritative state.	Place raster/raymarch/SDF preview in a replaceable adapter layer. Do not mistake the projector for the state space.	RETAIN	S3,S6, pp. 1-10;1-12
A12	Hybrid layer separation Information architecture	Persistent world state, local sensing and display remain distinct.	Use core/query/projection/hardware modules with explicit interfaces. Avoid polar-everything and raster-everything.	RETAIN	S5,S6, pp. 2-12;1-12
A13	Branch and chamber routing Information architecture	Events route state through branches/sheets/chambers.	Use finite enums and explicit routing tables. Avoid implicit magic routing.	RETAIN	S6,S7, pp. 5-12;23-31
A14	Uncertainty/confidence Information architecture	Verified outputs include error, confidence and calibration.	Attach numeric intervals/confidence and solver status to every event. Essential for real systems.	RETAIN	S5,S6, pp. 1-12;8-12
A15	Schema-bound packed data caution Information architecture	Raw packed bits are meaningless without layout metadata.	Version all bitfields/LUTs and ship schema with data. Useful contrast from the COBOL sections.	RETAIN	S2,S8,S9,S5, pp. 27-29;26-28;26-28;7
A16	Agent on shared relation substrate Information architecture	Perception/action can issue the same support/event queries.	Keep agent integration minimal: queries and transition choices, not universal AI compression. Do not claim an LLM is replaced by the geometry.	TRANSLATE	S6,S7, pp. 11-12;34-36
K01	Analytic motion component Game technology	Position/velocity/acceleration are closed-form trajectories.	Entity component stores trajectory coefficients and state patches. Supports horizon skipping for fixed formulas.	RETAIN	S6,S7,S4, pp. 7-12;26-31;10-16
K02	Continuous collision/event component Game technology	Collisions are relation roots under support/compatibility.	Event solver returns ordered crossings with guard semantics. Fallback numerical solver is allowed.	RETAIN	S6,S7, pp. 7-12;26-31
K03	Sensor cone component Game technology	Agent/sensor relevance uses local radial-angular supports.	FOV/range query before relation solving. Useful in Unity/Godot/ECS adapters.	RETAIN	S5,S6,S7, pp. 3-9;5-12;12-25
K04	Procedural grammar component Game technology	L-system/shape grammar generates bounded relation sets.	Compile grammar tokens to primitives, supports and transitions. Finite depth and normalization required.	RETAIN	S6,S7,S9, pp. 11-12;26-31;5-8
K05	Topological portal component Game technology	Mobius/Klein/hourglass routes connect spaces with orientation/sheet changes.	Portal has entry surface, exit transform, sheet map and lineage event. Can be rendered conventionally while simulated symbolically.	RETAIN	S5,S6,S7,S9, pp. 2-7;5-12;18-25;5-8
K06	Deterministic network/event replication Game technology	Seed plus event log reconstructs closed dynamics.	Replicate authoritative events and schema versions, not every visual frame. Only works where deterministic formulas are stable across platforms.	TRANSLATE	S6, pp. 8-12
K07	Graphics preview component Game technology	SDF/log-polar/jitter projects core state to pixels.	Optional GPU or CPU preview; no authority over simulation. Provides practical game-engine visualization.	RETAIN	S1,S3,S6, pp. 10-15;1-10;9-10

continued on next page

Table A.1 continued

ID	Category / name	Source expression	Normalized operator and qualification	Status	Source trace
H01	Bounded Compatibility Event (B.C.E.) Hardware endpoint	A verified event is declared only after support, compatibility and a measured guard crossing.	Reuse the core event schema for a measured optofluidic front end. The best-defined physical endpoint in the corpus.	RETAIN	S5, pp. 1-12
H02	Local support -> liquid lens -> mode -> guard Hardware endpoint	Radial-angular input is tuned, coupled, interfered and thresholded.	Separate optical, fluid, mode and digital-control models. Maxwell, Young-Laplace/coupled-mode theory and control logic stay type-separated.	RETAIN	S5, pp. 3-9
H03	Matrix-in-glass Hardware endpoint	A waveguide mesh implements a calibrated transfer function.	Treat as a measured passive/tunable transfer matrix, not stored imagery. Requires calibration and drift tracking.	RETAIN	S5, pp. 2-5
H04	One-bit hardware route flag Hardware endpoint	Parity denotes admission/route/freshness, not amplitude.	Store amplitudes, thresholds, uncertainty and lineage separately. Same type-safety rule as software.	RETAIN	S5, pp. 2-7
H05	Spherical throughput metric Hardware endpoint	Count verified events/queries per second at a declared error budget.	Report support, compatibility, miss/false rates, energy, latency and drift. Not photon flux alone.	RETAIN	S5, pp. 3,6,9-12
H06	Hollowlens-0 demonstrator Hardware endpoint	Scanned input, tunable liquid coupler, 2x2 interferometer, balanced detectors, digital sidecar.	Prototype only after guard, compatibility, calibration and baseline are preregistered. Can fail usefully.	RETAIN	S5, pp. 4,8-10

PART B

Reference world schema

B.1 Example world

The included JSON defines two entities, one line event surface, a local radial-angular support, sheet/phase/tag compatibility, and a transition that toggles sheet, flips orientation, changes branch and adds phase.

Listing B.1: UGTS-0 example world

```
1 {
2   "schema_version": "1.0.0",
3   "metadata": {
4     "description": "Minimal double-vacuum and parity-hinge demonstration."
5   },
6   "entities": [
7     {
8       "id": "traveler_A",
9       "trajectory": {
10        "type": "linear",
11        "p0": [
12          -2.0,
13          0.25
14        ],
15        "v0": [
16          1.0,
17          0.0
18        ],
19        "t0": 0.0
20      },
21      "phase": 0.0,
22      "sheet": 0,
23      "orientation": 1,
24      "branch": "A",
25      "tags": [
26        "player"
27      ]
28    },
29    {
30      "id": "co_located_B",
31      "trajectory": {
32        "type": "linear",
33        "p0": [
34          0.0,
35          0.25
36        ],
37        "v0": [
38          0.0,
39          0.0
40        ],
41        "t0": 0.0
42      },
43      "phase": 3.141592653589793,
44      "sheet": 1,
45      "orientation": -1,
46      "branch": "B",
47      "tags": [
48        "ghost"
49      ]
50    }
51  ],
52  "relations": [
```

```

53   {
54     "id": "x_zero_hinge",
55     "surface": {
56       "type": "line",
57       "normal": [
58         1.0,
59         0.0
60       ],
61       "offset": 0.0
62     },
63     "support": {
64       "type": "radial_angular",
65       "origin": [
66         0.0,
67         0.0
68       ],
69       "r_min": 0.0,
70       "r_max": 3.0,
71       "theta_center": 0.0,
72       "theta_half_width": 3.141592653589793
73     },
74     "compatibility": {
75       "allowed_sheets": [
76         0
77       ],
78       "phase_center": 0.0,
79       "phase_tolerance": 0.2,
80       "required_tags": [
81         "player"
82       ]
83     },
84     "transition": {
85       "toggle_sheet": true,
86       "flip_orientation": true,
87       "set_branch": "B",
88       "phase_delta": 1.5707963267948966
89     }
90   }
91 ],
92 "topology": [
93   {
94     "type": "mobius_band",
95     "width": 4.0,
96     "height": 2.0
97   }
98 ]
99 }

```

B.2 Schema design rules

1. Every bitfield has a declared meaning and schema version.
2. Position, time, phase, sheet, orientation, branch, lineage and uncertainty are distinct.
3. Surface solver family is declared and logged.
4. Projection settings are not authoritative simulation state.
5. External novelty and calibration changes enter the event log.

PART C

Source notes index

Per-document notes are shipped under `source_notes/`. They identify page-level mechanisms, retained operators and excluded extrapolations. The source PDFs themselves are not copied into the ZIP.

Ref	Note file	Primary contribution
S1	S1_NUM19.md	Radix thresholds, Hamming weight, triangle seed, Pascal parity, fractal and graphics/topology motif chain.
S2	S2_AUTHOR_DIALOGUE.md	Duplicate glyph/topology thread plus schema, AR and speculative extensions.
S3	S3_KC_GRAPHICS.md	Log-polar/SDF/one-bit graphics and prepress adapter.
S4	S4_BEN_BURGER.md	Zero indexing, split oval and shared overlap lens.
S5	S5_WAVEGUIDE.md	B.C.E., optofluidic signal path, metrics and kill criteria.
S6	S6_CHRONO_SYNTHESIS.md	Mature query-first event calculus and feasibility discipline.
S7	S7_HOLLOWLAND.md	Double vacuum, phase sheets, hourglass, grammar and grand claims.
S8	S8_GEORGE_BUNDLE.md	Overlapping dialogue bundle; deduplicated.
S9	S9_D_T0_PH.md	Glyph morph, parity hinge, log-polar/L-system and SDF-global morph.

Closing statement

The corpus's most valuable idea survives only after discipline: geometry defines relations and local supports; topology defines explicit gluing and compatibility sectors; information architecture defines typed state, lineage and event semantics; game technology supplies components and adapters; hardware supplies measurable guards and error budgets. With those boundaries, the material becomes a coherent substrate that can be implemented, tested and falsified.